

*TN5A**
Ameritrade

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Objectives

- Ameritrade Holding Corporation (AMTD) is a deep-discount brokerage firm that has recently completed an IPO.
 - Opportunity to estimate the cost of capital.
- Explore how capital market data and the CAPM can be used to estimate the required rate of return for **real** investments.
- The case will help us to cement ideas about discount rates:
 - Determine which firms can be considered **comparable** or **pure play** when estimating the cost of capital.
 - Estimate equity and asset betas from capital market data.
 - Evaluate the appropriateness of the estimated cost of capital for different types of real investments.

Evaluating Investments in Technology

- AMTD's potential investments are expected to yield cash flows that will accrue over the next several years.
- AMTD is planning to spend:
 - \$75M in advertising for 1998.
 - \$100M to upgrade its technology and physical facilities.
- These investments are **large** relative to AMTD's book and market value of equity.
 - Book Value of Equity = **\$66.9M** [Ex.2]
 - Market Value of Equity = **\$164M – \$273M** [Ex.5]
- **Bottom Line:** When compared to market equity, the relative size of AMTD's planned investment is very large.

Cost of Capital at Ameritrade

- A firm maximizes its value by taking all positive NPV projects.

$$V = E(CF_0) + \frac{E(CF_1)}{1 + r_A} + \frac{E(CF_2)}{(1 + r_A)^2} \dots$$

- We need to look for the unlevered cost of capital, r_A .
 - Investors require compensation for systematic **business risk**
 - **Need to follow the “five step” recipe (on slide 11)**

Using CAPM to Estimate the Cost of Capital

- The CAPM provides a useful framework for determining the discount rate of the project.
 - Investors get compensation for bearing the market risk of the project.
 - Pure **business risk** compensation!

$$E[r_A^{\text{Ameritrade}}] = r_f + \beta_A^{\text{Ameritrade}} \times [E(r_m) - r_f]$$

- We need to find or estimate each of the inputs to the CAPM:
 - Risk-Free Rate
 - Market Risk-Premium
 - *Asset* Beta for Ameritrade

Inputs for CAPM: Risk-Free Rate

- **Definition:** Rate of return on future cash flows that are certain.
- **Match** the risk-free rate's maturity to the economic life of the project.
 - This is a long-term project since benefits from the investment will be seen in the future.
- You need to use the prevailing yields in the market on US Government Securities [**Ex. 3**].

Inputs for CAPM: Market Risk Premium

- **Definition:** Expected return of the market in excess of the risk-free rate required to compensate investors for risk of uncertain future cash flows.
- Issues to discuss when choosing the market risk premium:
 - The market portfolio in the CAPM includes **all** risky assets, not just stocks.
 - Use the stock market return *minus* the US government bond return as a **proxy**.
 - Reliability of historical data: Should we use long or short-term series?
 - Long series are better. Used 1929-1996!
 - Large stocks versus small stocks?
 - Large stock portfolio is better estimate of the market portfolio. Why?

Inputs for CAPM: Market Risk Premium (cont.)

- Since we do not have expected returns, use long-term historical returns on **both** market returns and risk-free rate.
 - A sensible approach is to use the average annual return of large stocks minus the average annual US government bond return over the longer time period. $12.7\% - 5.5\% = 7.2\%$.

Inputs for CAPM: Asset Beta

- **Definition:** Measure of systematic risk of the expected cash flows of the project. Compensation only for business risk!
- Since AMTD has almost **no** public information available (*i.e.* public since March 1997), we need to use the “Twin Method” or “Comparable Method” to unlever and relever.
- Let’s review very quickly the unlevering process we studied before.

Recall: Market Value Balance Sheet

- We are interested in calculating the $E(r_A)$, but generally we cannot observe market value asset returns
 - Since the balance sheet must balance at each point in time, the expected return on the LHS must equal the expected return on the RHS.

<u>Assets</u>	<u>Liabilities</u>
Equip. & Plant (A)	Debt (D) Equity (E)
	↓
	A = D + E

$$E[r_A] = \frac{D}{D + E} E[r_D] + \frac{E}{D + E} E[r_E] \Rightarrow E[r_A] = \frac{D}{A} E[r_D] + \frac{E}{A} E[r_E]$$

- The CAPM can price any type of security:

$$E[r_D] = r_f + \beta_D \times (E[r_m] - r_f); \quad E[r_E] = r_f + \beta_E \times (E[r_m] - r_f)_{t_0}$$

Market Value Balance Sheet (cont.)

- Substituting the CAPM equations for $E(r_D)$ and $E(r_E)$ we obtain:

$$\beta_A = \frac{D}{D+E} \beta_D + \frac{E}{D+E} \beta_E \Rightarrow \beta_A = \frac{D}{A} \beta_D + \frac{E}{A} \beta_E$$

- We need to calculate β_A using the comparable approach.
 - Recognize that all firms that generate cash flow streams with the same risk will have the **same** β_A .
 - **Step 1:** Find an identical “twin” or comparable that matches the project’s characteristics. You do not care about financial risk!
 - **Step 2:** Find their β_E . Regress stock returns on market returns.
 - **Step 3:** Get their β_A by “unlevering” their β_E .
 - Using the *actual/historical* market values of D and E for each of the twins.
 - **Step 4:** Take an average of the β_A ’s.
 - **Step 5:** Apply CAPM

Step 1: Looking For Comparables

- The assumption we are making when identifying comparable firms is that the riskiness of the cash flow streams of the different firms is the **same**.
 - We expect business risk to be the same independent of the capital structure. By unlevering, we are eliminating the financial risk from the firm and keeping only the business risk.
- **Exhibit 4** displays **three** types of firms:
 - Investment Services
 - Includes brokerage, asset management, IB, and trading.
 - Discount Brokerages
 - Internet
- **Bottom Line:** Ameritrade is a **discount broker** deriving its revenues from brokerage commissions and net interest revenues.

Step 1: Looking For Comparables (cont.)

- The sources of revenues for Ameritrade are **heavily dependent** on overall stock market activity.
- Full service firms such as Merrill Lynch have asset management, investment banking, and trading in addition to brokerage.
 - Asset management and IB activities may be **less** correlated with the overall stock market than basic brokerage.
- Ameritrade is starting to use the Internet to facilitate self-directed trading, but is **not** an Internet firm.
- **Bottom Line:** E*Trade, Charles Schwab, Quick & Reilly, and Waterhouse Investor Service seem to be the **most reasonable** comparable firms.
 - At least **80%** of their revenues come from brokerage [**Ex.4**].

Step 2: Find β_E of Comparable Firms

- I will use as **comparable** firms:
 - Charles Schwab, Quick & Reilly, and Waterhouse Investor Service.
 - Why not E*Trade?
 - E*Trade has also recently gone public and has limited capital market data.
- To estimate β_E , we first need to compute stock returns from the price and dividend data of these 3 companies.
 - We need to be careful with the stock splits!
- **Question:** What are stock splits?

Side Note: What are Stock Splits?

- **Stock Split** refers to the corporate action that increases the number of shares in a public company.
- The price of the shares is adjusted so that the pre-split and post-split market capitalization of the company remains the same and dilution does not occur.
- **Example:** A company has 100 shares of stock, each with a price of \$50. The market capitalization is $100 \times \$50 = \$5,000$.
 - The company splits its stock “2-for-1”. There are now 200 shares of stock and each shareholder holds twice as many shares. The price of each share has been adjusted to \$25. The market capitalization is $200 \times \$25 = \$5,000$, the **same** as before the split.
- Ratios of “2-for-1”, “3-for-1”, and “3-for-2” splits are the most common but any ratio is possible.

Side Note: What Are Stock Splits? (cont.)

- It is often claimed that stock splits, in and of themselves, lead to higher stock prices; however, research does **not** bear this out. What is true is that stock splits are usually initiated after a large run up in share price.
 - Momentum investing would suggest that such a trend would continue regardless of the stock split.
- Other effects could be psychological.
 - If many investors think that a stock split will result in an increased share price and therefore purchase the stock, the share price will indeed tend to increase.
- Others contend that the management of a company, by initiating a stock split, is implicitly conveying its confidence in the future prospects of the company.

Calculating Historical Returns on Equity

- Need to be careful with the time period chosen to compute historical returns on equity.
 - Although for the 3 comparables we have data beyond 1992, we will focus on the time period **1992-1996**!
 - Given the data provided in the case [Ex.4], it makes sense to estimate regressions over 5-year periods that **correspond** to the capital structure provided.
- **Remark:** It is very important to match the available capital structure data for the firms to the estimation period, as leverage will have an effect on equity betas!
- To compute historical returns, use the following equations for “if no stock split” and “if stock split”, respectively:

$$r_t = \frac{P_t - P_{t-1} + D_t}{P_{t-1}}; \quad r_t = \frac{\frac{x}{y}P_t - P_{t-1} + \frac{x}{y}D_t}{P_{t-1}}$$

where x/y is the ratio of new shares outstanding to old shares outstanding. For example, a “2 for 1” split means $x/y = 2/1$.

Step 2: Calculating β_E

- To calculate β_E for each of the comparable firms, you can either:
 - Perform a **regression** of the **market model** for each of the returns of the comparables against the **value-weighted market return**.

$$E(r_i) = \alpha_i + \beta_i r_m + \varepsilon_i$$

- Use the statistical definition of beta:

$$\beta_{E,i} = \frac{Cov(r_{E,i}, r_m)}{Var(r_m)}$$

- **Note:** Obviously both ways should yield the **same** β_E .

Step 2: Calculating β_E (cont.)

- As expected, both methods yield exactly the **same** results:

Charles Schwab (1/92-12/96)

Stock	Market
Average Return 3.345%	Average Return 1.182%
Std. Dev. 11.513%	Std. Dev. 2.476%
Variance 1.325%	Variance 0.061%
Covariance 0.141%	Equity Beta 2.297
Regression Beta	2.297

Quick & Reilly (1/92-12/96)

Stock	Market
Average Return 2.339%	Average Return 1.182%
Std. Dev. 10.397%	Std. Dev. 0.024764681
Variance 1.081%	Variance 0.061%
Covariance 0.135%	Equity Beta 2.204
Regression Beta	2.204

Waterhouse (1/92-9/96)

Stock	Market
Average Return 4.117%	Average Return 1.125%
Std. Dev. 16.348%	Std. Dev. 2.417%
Variance 2.673%	Variance 0.058%
Covariance 0.186%	Equity Beta 3.190
Regression Beta	3.190

Step 3 & 4: Computing β_A by Unlevering β_E and taking averages

Comparable Firms	Debt/Value (Market Value)	Equity Beta	Unlevering Process	
			Asset Beta (Assume $\beta_D = 0$)	Asset Beta (Assume $\beta_D = 0.25$)
Charles Schwab	0.08	2.30	2.11	2.13
Quick & Reilly	0.00	2.20	2.20	2.20
Waterhouse	0.38	3.19	1.98	2.07
Equally Weighted Avg.			2.10	2.14

Investment Service Firms	Debt/Value (Market Value)	Equity Beta	Unlevering Process	
			Asset Beta (Assume $\beta_D = 0$)	Asset Beta (Assume $\beta_D = 0.25$)
A G Edwards	0.00	1.45	1.45	1.45
Bear Sterns	0.50	1.69	0.85	0.97
Merril Lynch	0.52	1.82	0.87	1.00
Morgan Stanley	0.53	2.09	0.98	1.11
Paine Webber	0.53	1.76	0.83	0.96
Raymond James Financial	0.04	1.45	1.39	1.40
Equally Weighted Avg.			1.06	1.14

- **Remark I:** Note that the β_A seem approximately the **same**.
- **Remark II:** The relation between leverage and equity beta shows up clearly! Higher D/V is reflected in **higher** β_E !

Step 5: Using CAPM

Cost of Capital Estimates

For Discount Brokers and Investment Service Firms

$$E(r_A) = r_f + \beta_A \times [E(r_m) - r_f] = .0661 + \beta_A \times 0.072$$

Type Of Business	r_f	β_A	$r_m - r_f$	CAPM $r_A (\beta_D = 0)$	CAPM $r_A (\beta_D = 0.25)$
Discount Brokerage	0.0661	2.10	0.072	21.7%	22.0%
Investment Services	0.0661	1.06	0.072	14.2%	14.8%

- The 21-22% cost of capital estimate represents the risk of real investments in the discount brokerage industry.
- The relatively high cost of capital is largely **unavoidable**.
 - This business has a very high systematic risk since all revenues are directly linked to stock market activity and performance.

What Happened?

- Ameritrade increased its competitive position by engaging in the technology investments.
 - Ameritrade was known for its “trade execution speed” during the early 2000s.
 - Using Ameritrade’s stock returns from the IPO in 1997 through 2000, a direct cost of capital estimate can be calculated.
 - The $\beta_A = 2.1$ is very much in line with the new estimate of **2.3**!

Firm	Debt/Value (Market Value)	Equity Beta	Unlevering Process	
			Asset Beta (Assume $\beta_D = 0$)	Asset Beta (Assume $\beta_D = 0.25$)
Ameritrade	0.14	2.71	2.32	2.36

Why This Case?

- Explored the alternatives for computing the cost of capital for a traded public company with not enough information.
 - Opportunity Cost of Capital
 - We care about the “business risk”
 - Depends on: risk of project (measured by beta); market price of risk (risk free rate and risk premium)
- Estimated cost of capital:
 - Used “Twin or Comparable Method”.
 - Calculated β_E for comparables using regression analysis.
 - Unlevered β_E to get β_A .
 - Computed the cost of capital using CAPM.